

Cambridge International AS & A Level

CANDIDATE
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FURTHER MATHEMATICS

9231/22

Paper 2 Further Pure Mathematics 2

October/November 2023

2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

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- 1 Find the Maclaurin's series for $\ln(x+2) + \ln(x^2+5)$ up to and including the term in x^2 . [5]

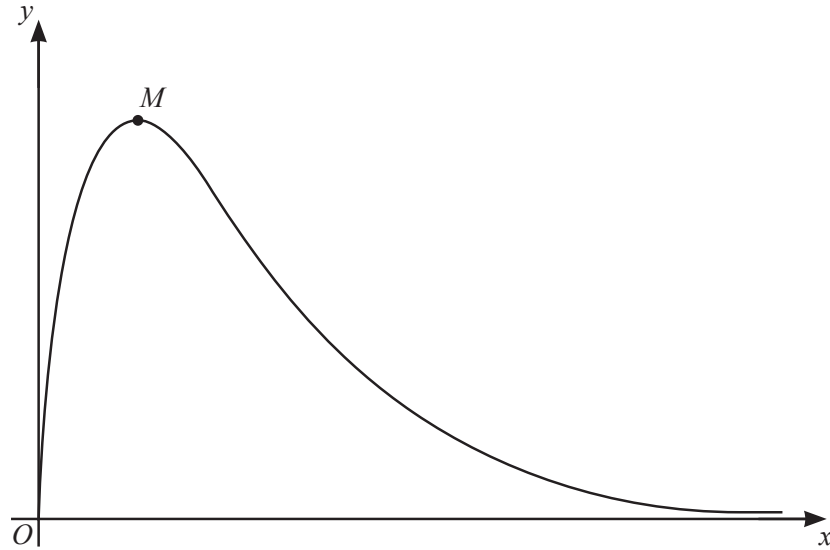
[illegible]

- 4 Find the solution of the differential equation

$$\frac{dy}{dx} + 3y = \sin x$$

for which $y = 1$ when $x = 0$. Give your answer in the form $y = f(x)$. [9]

[illegible]



The diagram shows part of the curve $y = x \operatorname{sech}^2 x$ and its maximum point M .

- (a) Show that, at M ,

$$2x \tanh x - 1 = 0$$

and verify that this equation has a root between 0.7 and 0.8. [4]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

(b) By considering a suitable set of rectangles, use the diagram to show that

$$\sum_{r=2}^n r \operatorname{sech}^2 r < n \tanh n + \ln \operatorname{sech} n - \tanh 1 - \ln \operatorname{sech} 1. \quad [6]$$

The 3×3 matrix $\mathbf{A}$ has distinct non-zero eigenvalues $a, \frac{1}{2}, 2$ with corresponding eigenvectors

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix},$$

respectively.

(c) Find $\mathbf{A}^{-1}$ in terms of a .

[5]

[illegible]

This image shows a full page of white paper with ten horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and extend across the width of the page. There is no text or other markings on the paper.

- Use the substitution $x = \frac{1}{2} \sinh u$ to show that $S = \frac{1}{32} \pi \left(\frac{820}{81} - \ln 3 \right)$. [9]

[illegible]

$$y^3 \frac{d^2 y}{dx^2} + 3y^2 \left(\frac{dy}{dx} \right)^2 + y^3 \frac{dy}{dx} + y^4 = e^{-2x}.$$

$$\frac{d^2v}{dx^2} + \frac{dv}{dx} + 4v = 4e^{-2x}. \quad [4]$$
This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- (b)** Find y in terms of x , given that, when $x = 0$, $y = 1$ and $\frac{dy}{dx} = -\frac{3}{8}$. [10]

[illegible]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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